

Cells and Organs of the Immune System

Hematopoiesis

- The process that refers to the formation and development of cells of the blood
- Requires a unique progenitor cell:
 - ✓ The hematopoietic stem cell (HSC)
- Hematopoiesis occurs in the bone marrow
- Sites of Hematopoiesis
 - **Fetus**
 - 0-2 months yolk sack
 - 2-7 months liver and spleen
 - 5-9 months bone marrow
 - **Infants** **Bone marrow (all sites)**
 - **Adults** Bone marrow: vertebrae, ribs
sternum, skull, sacrum and pelvis, proximal femur

Hematopoietic Stem Cells

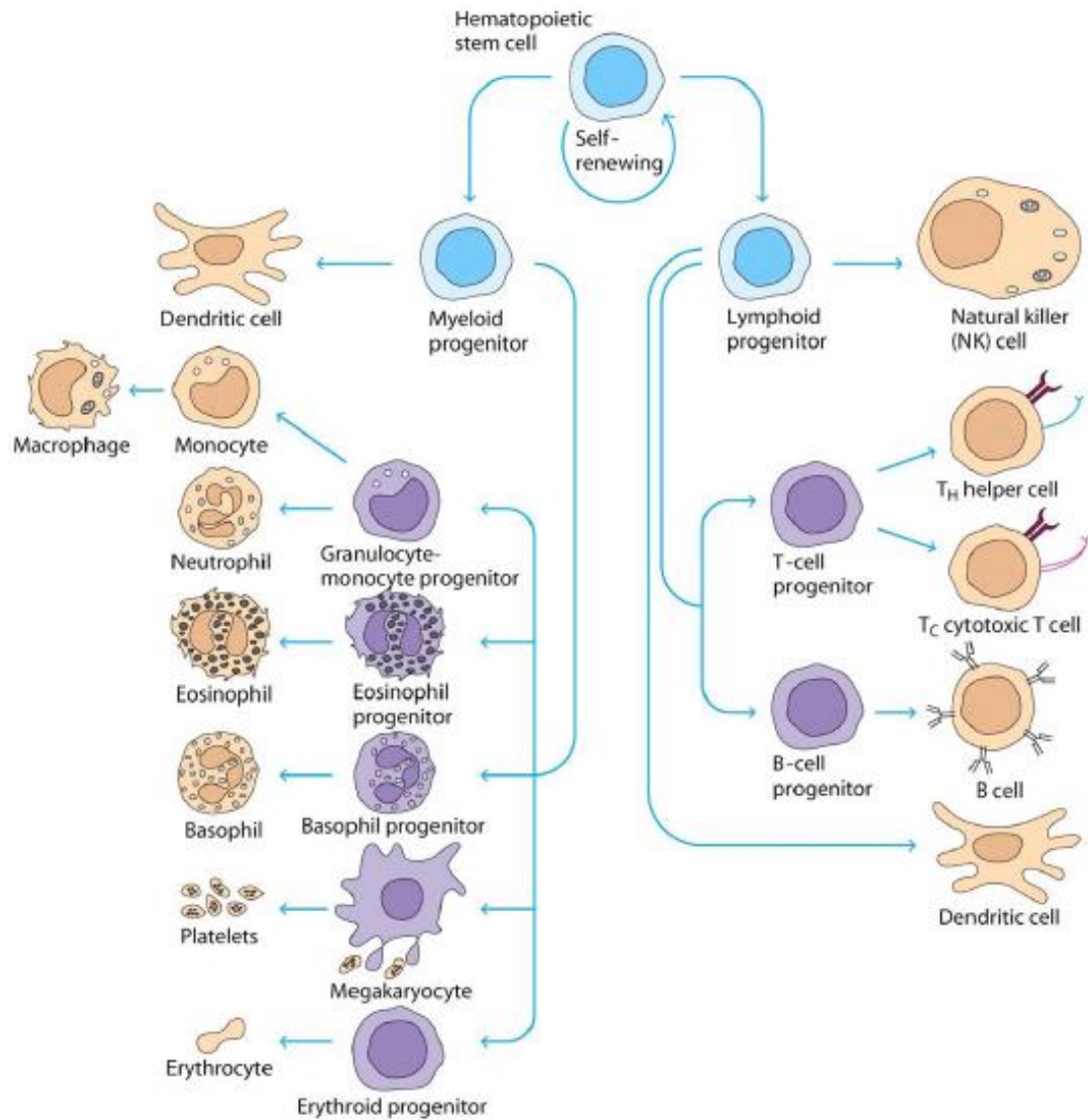
- HSCs have incredible proliferative capacity
- HSCs are very rare: 1 in 50,000 bone marrow cells
- HSCs are capable of self renewal

1. Self renewal, ie producing more HSCs, and
2. Differentiating into all of the cells found in the blood

=pluripotent

Pluripotent Stem differentiates into

- Myeloid Stem Cell Progenitor
- Lymphoid Stem Cell Progenitor



Regulation of Hematopoiesis

- In bone marrow hematopoietic cells grow & mature on a meshwork of **stromal cells**
 - which are nonhematopoietic cells
 - support growth & differentiation
 - include
 - fat cells, endothelial cells, fibroblasts & macrophages

Regulation of Hematopoiesis

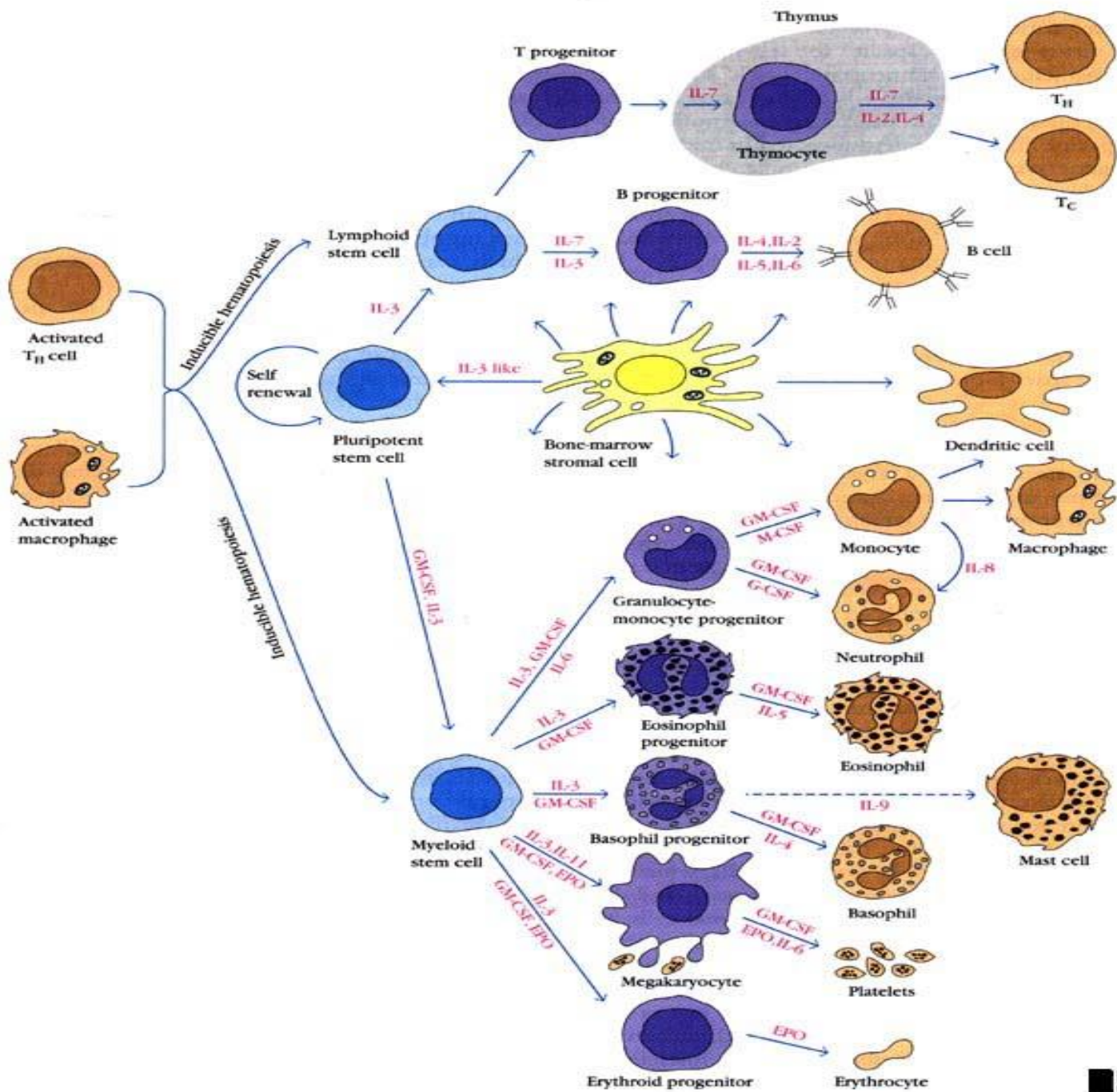
- influence differentiation of hematopoietic stem cells
by providing a hematopoietic-inducing
microenvironment (HIM)
- consisting of a cellular matrix & factors that promote
growth & differentiation

Regulation of Hematopoiesis

- progenitor commitment depends on—acquisition of responsiveness to particular growth factors & cytokines
 - Multilineage CSF or interleukin-3
 - Macrophage CSF
 - Granulocyte CSF
 - granulocyte-macrophage CSF

- CSFs- act in a stepwise manner inducing proper maturation.
 - IL-3 [multi-CSF] acts early, possibly to induce formation of the nonlymphoid cells
 - GM-CSF acts at a slightly later stage, but it also induces formation of all the nonlymphoid blood cells.
 - M-CSF and G-CSF act still later to promote the formation of monocytes and granulocytic cells, respectively.
 - IL-4 - stimulates B progenitors, mast progenitors, and basophil progenitors
 - IL-5 - stimulates eosinophil progenitor
 - IL-6 - stimulates the myeloid stem cell
 - IL-7 - induces the differentiation of lymphoid progenitor into B progenitor and T progenitor
 - IL-8 - stimulates the neutrophil progenitor
 - IL-9 - stimulates mast cell growth

- Commitment of a progenitor cell is associated with the expression of receptors on the cell membrane that are specific for particular cytokines.
- Hematopoiesis is a continuous process throughout adulthood. Production of mature blood cells equals their loss. Estimated that the average human must produce 3.7×10^{11} blood cells per day.
- Cell division and differentiation during hematopoiesis are balanced by apoptosis



Cells of the Immune System

- The immune system has five types of cells that perform all basic defence functions. From those, **the lymphocytes** are the most important.
- Blood contains three major components:
 1. Plasma or Serum
soluble factors: complement, antibodies, etc
 2. White blood cells = leukocytes
Lymphocytes, Macrophages, Neutrophils, etc.
=Cells of the immune system: Innate and adaptive
 3. Red blood cells
Platelets, Red blood cells

TABLE 2-4**Normal adult blood-cell counts** $=10^{-6}$ Liters**Cell type****Cells/mm³****%**

Red blood cells

 5.0×10^6 $=2.5 \times 10^{13}$ in an adult

Platelets

 2.5×10^5

Leukocytes

 7.3×10^3 $=3.65 \times 10^{10}$ in an adult

Neutrophil

50–70

Lymphocyte

20–40

Monocyte

1–6

Eosinophil

1–3

Basophil

<1

A. Lymphoid Cells

- Lymphocytes normally represent 20-40% of the WBCs
- Lymphocytes are responsible for the two defining characteristics of the adaptive immune response, **specificity and memory**
- Lymphocytes circulate back and forth between the blood and the lymphoid system of the body.
- They have a life span of days to years.
- There are 3 major populations of lymphocytes

B Lymphocytes:

- In mammals, B cells mature in the bone marrow.
- The hallmark of B lymphocytes is the synthesis and display of antibody (or immunoglobulins) in their cell membranes.
- All antibody molecules on a single B-cell have an identical binding site for antigen

T Lymphocytes:

- Generally 60-80% of the lymphocytes are T-lymphocytes
- These cells mature in the thymus.
- Like the B-cells, they express on their membrane an antigen-binding receptor called T-cell receptor or TCR. These receptors do not recognize the antigen directly. They recognize the antigen bound to a molecule of the Major Histocompatibility Complex (MHC).

- There are two populations of T-cells, T helper (TH) and T cytotoxic (TC).
 - TH cells express the marker CD4 and recognize antigens bound to MHC class II molecules.
 - TC cells express the marker CD8 and recognize antigens bound to MHC class I molecules

NK cells (natural killer cells)

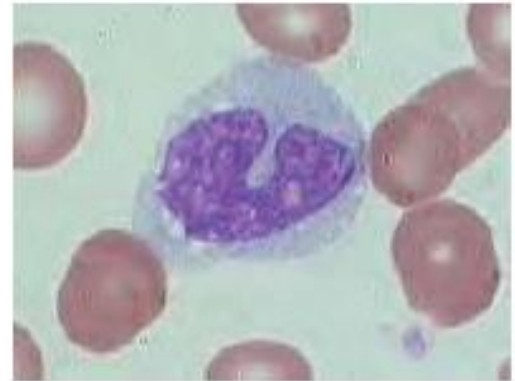
- Are large granular lymphocyte that lack B-cell receptors and T-cell receptors.
- NK cells represent a first line of defense to infections, tumor growth and other pathogenic alterations of tissue homeostasis.
- NK cells produce cytokines and express receptors for immunoglobulin.

Lymphocyte Classes

Class	Functions	Antigen Receptor	Selected Phenotypic Markers	Percent of Total Lymphocytes		
				Blood	Lymph Node	Spleen
B lymphocytes	Antibody production (humoral immunity)	Surface antibody (immunoglobulin)	Fc receptors; class II MHC; CD19; CD21	10–15	20–25	40–45
T lymphocytes						
Helper T lymphocytes	Stimuli for B cell growth and differentiation (humoral immunity) Macrophage activation by secreted cytokines (cell-mediated immunity)	$\alpha\beta$ heterodimers	CD3 ⁺ , CD4 ⁺ , CD8 ⁻	50–60*	50–60	50–60
Cytolytic T lymphocytes	Lysis of virus-infected cells, tumor cells, allografts (cell-mediated immunity)	$\alpha\beta$ heterodimers	CD3 ⁺ , CD4 ⁻ , CD8 ⁺	20–25	15–20	10–15
Natural killer cells	Lysis of virus-infected cells, tumor cells; antibody-dependent cellular cytotoxicity	?	Fc receptor for IgG (CD16)	~10	Rare	~10

B. Mononuclear Cells

- Mononuclear cells are the monocytes and macrophages. Promonocytes leave the bone marrow and differentiate into circulating monocytes. Monocytes circulate in the blood for about 8 hrs, and then migrate to the tissues and differentiate into tissue-specific macrophages.
- Macrophages have different names depending on their location in the body:
 - Alveolar macrophages: Lungs
 - Histocytes: Connective tissue
 - Kupffer cells: Liver
 - Mesengial cells: Kidney
 - Microglial cells: brain
 - Osteoclast: bone



C. Granulocytic Cells

There are three types of Granulocytes:

- Neutrophils,
- Eosinophils
- Basophils.

Neutrophils

- Neutrophils are the most abundant of the leukocytes, normally accounting for 50-70% of the WBCs.
- Their granules contain various agents for killing microbes. These include lysozyme, lactoferrin, acid hydrolase, and myeloperoxidase.
- They also express receptors for immunoglobulin and complement and are involved in the acute inflammatory response.

Eosinophils

- Eosinophils normally comprise 1-4% of the WBCs
- Their granules contain destructive enzymes such as acid phosphatase, peroxidases, and proteinases for killing infectious organisms.
- They are also capable of phagocytosis but primarily they release their contents into the surrounding environment to kill microbes extracellularly.
- Eosinophils carry receptors for IgE, are involved in the destruction of IgE coated parasites, such as helminths, and contribute to the response to allergens.

Basophils

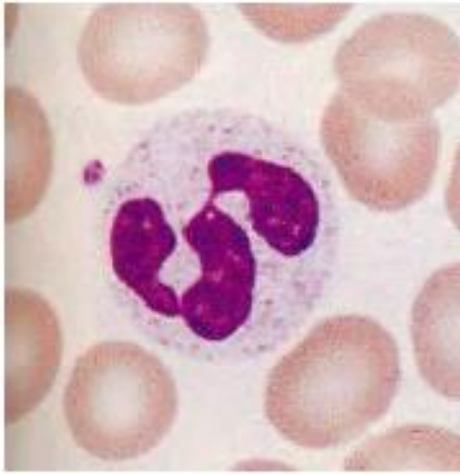
- Basophils normally make up 0-1% of the WBCs
- Non-phagocytic
- Basophils release histamine, leukotrienes, and prostaglandins, chemicals that promotes inflammation
- They express high affinity receptors for IgE and are stimulated to secrete the chemicals responsible for immediate hypersensitivity

D. Mast Cells

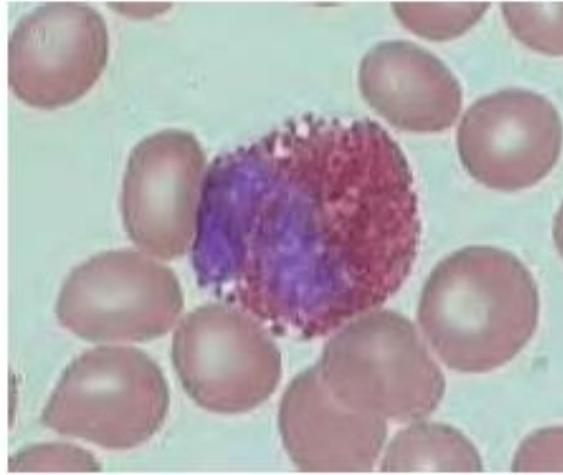
- They are found in a wide variety of tissues and are characterized by having granules containing histamine.
- Together with basophils are important in the development of allergies.

E. Dendritic Cells

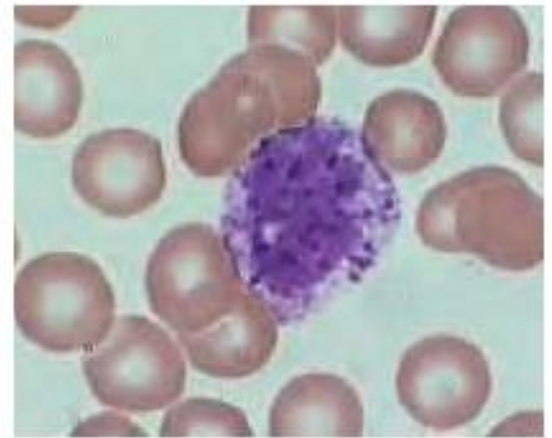
- They have membrane extensions that resemble those of nerve cells.
- They are antigen-presenting cells and are classified according to their location into:
 - Langerhans cells: in the skin
 - Interstitial dendritic cells: In connective tissue of most organs
 - Interdigitating dendritic cells: In lymphoid tissues
 - Circulating dendritic cells
 - Follicular dendritic cells: in Lymph nodes.



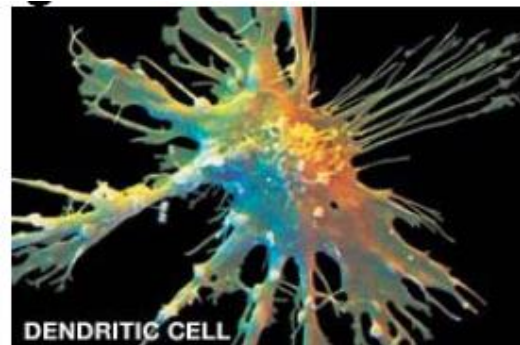
Neutrophils



Eosinophils



Basophiles



Organs of the Immune system

Primary lymphoid organs-

- where maturation takes place
- where lymphocytes become immunocompetent
- Thymus and bone marrow
 - Thymus= T-cells
 - Bone marrow= B-cells

Secondary lymphoid organs-

- where lymphoid cells encounter antigen
- where lymphoid cells are activated
- Lymph nodes, Spleen, Cutaneous and Mucosal associated lymphoid tissue

Bone Marrow

A site of generation of all circulating blood cells in the adult, including immature lymphocytes , and is the site of B cell maturation

All the blood cells originate from a common stem cells that differentiate into a particular lineages (i.e. erythroid, megakaryocytic, granulocytic, monocytic, and lymphocytic)

It also harbors numerous antibody secreting plasma cells

Thymus

- It is the site of T cell maturation
- Immature cells of T cell lineage enter the thymus , start maturation in the thymic cortex, migrate to the medulla and when maturation is completed leave the thymus

Lymph nodes

Lymph nodes are small nodular aggregates of lymphocyte-rich tissue situated along lymphatic channels throughout the body

Site for the generation of adaptive immune responses to lymph-borne antigens

Different classes of lymphocytes are sequestered in distinct regions of lymph nodes, where follicles are B cell zones and cortex serves as a T cell zone

Spleen

A major site for generation of immune response to blood-borne antigens

- Lymphocytes and accessory cells are anatomically segregated in the spleen
- It also serves as an important filter for the blood as a result of macrophage in the red pulp

Cutaneous immune system

The skin contains a specialized cutaneous immune system consisting of lymphocytes and accessory cells.

- The principal cell populations within the epidermis are keratinocytes, melanocytes, langerhans cells and intraepithelial

T cells

- langerhans cells 1% of cells in the epidermis but cover 25% of the surface
- Intraepidermal lymphocytes constitute only about 2% of skin associated lymphocytes and the majority are CD 8 cells

Mucosal Immune System

- Mucous membranes cover the digestive, respiratory, and urogenital tracts.
- They are the major site of entry for most pathogens. Hence, these sites are protected by a group of lymphoid tissue known as mucosal-associated lymphoid tissue (MALT).
- The largest mammalian lymphoid organ system, and comprises about 80% of all lymphocytes

Mucosal Immune System

- Plasma cells in MALTs exceed in number those found in lymph nodes, spleen, and bone marrow combined.
- Well organized as the appendix and Peyer's patches, or Barely organized as those found in the lamina propria of the intestinal epithelium.

Mucosal Immune system

- **Subdivisions of MALT:**
 - **GALT** (GI tract)
 - **BALT** (bronchial tree)
 - **NALT** (nasopharyngeal area)
 - Mammary gland, salivary and lacrimal glands, genitourinary organs

- The continuous movement of lymphocytes via the blood stream and lymphatics, from one peripheral lymphoid tissue to another, and then to peripheral sites, is essential for effective immune responses to foreign antigens
- The patterns of lymphocyte recirculation are governed by the expression of adhesion molecules on lymphocytes and vascular endothelial cells

Reading Assignment

- Distribution of lymphocyte subsets in different anatomical regions of LN and spleen